HOMEWORK1:

1.**SUM OF TWO MATRICES:**

```
#include <stdio.h>
int main()
{
 int i,j,n,m;
printf("Enter the no of rows:");
scanf("%d",&n);
printf("Enter the no of columns:");
scanf("%d",&m);
int a[n][m],b[n][m],c[n][m];
if(n==m)
{
   printf("Enter the elements of first matrix:");
   for(i=0;i<n;i++)
   {
      for(j=0;j<m;j++)
      scanf("%d",&a[i][j]);
   }
   printf("Enter the elements of second matrix:");
   for(i=0;i<n;i++)
   {
```

```
        for(j=0;j<m;j++)
        scanf("%d",&b[i][j]);
    }
    printf("The resultant matrix is:");
    for(i=0;i<n;i++)
    {
        for(j=0;j<m;j++)
{
        c[i][j]=a[i][j]+b[i][j];
        printf("%d\t",c[i][j]);
    }


    printf("\n");

}
    else
    printf("Matrix addition is not possible");

return 0;
}
```

OUTPUT

Enter the no of rows:2

Enter the no of columns:2

Enter the elements of first matrix:1

2

3

4

Enter the elements of second matrix:1

2

3

4

The resultant matrix is:

2 4

6 8

2.REVERSE OF AN ARRAY

```
#include <stdio.h>


int main() {
    int n;
    printf("Enter the no of numbers:");
    scanf("%d",&n);
    int a[n],i;
    printf("Enter the numbers:");
```

```
    for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
    printf("The numbers in reverse order:");
    for(i=n-1;i>=0;i--)
    {
        printf("%d ",a[i]);
    }


    return 0;
}
```

OUTPUT

Enter the no of numbers:5

Enter the numbers:1 12 3 4 6

The numbers in reverse order:6 4 3 12 1

3.DUPLICATE ELEMENTS IN AN ARRAY

```
#include <stdio.h>
```

```
int main() {
    int n,duplicate=0;
    printf("Enter the no of numbers:");
    scanf("%d",&n);
    int a[n],i,j;
    printf("Enter the numbers:");

    for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
    for(i=0;i<n;i++)
    {

for(j=i+1;j<n;j++){
if(a[i]==a[j])
duplicate++;
break;
    }
}
    printf("Total no of dupicates=%d",duplicate);

    return 0;
}
```

OUTPUT

Enter the no of numbers:6

Enter the numbers:1

1

2

3

4

4

Total no of dupicates=2

4.UNIUE ELEMENT IN AN ARRAY

```
#include <stdio.h>

int main() {
    int n,duplicate=0;
    printf("Enter the no of numbers:");
    scanf("%d",&n);
    int a[n],i,j;
    printf("Enter the numbers:");

    for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
```

```
    printf("UNIQUE ELEMENTS ARE:");
    for(i=0;i<n;i++)
    {
   int count=0;
for(j=0;j<n;j++){
if(a[i]==a[j])
count++;

    }
if(count==1)
{
    printf("%d",a[i]);
}
}

    return 0;
}
```

OUTPUT

Enter the no of numbers:3

Enter the numbers:1

2

2

UNIQUE ELEMENTS ARE:1

**----------

5.Insert an Element in a Sorted Array**

```
#include <stdio.h>

int main()
{
   int a[100], n, i, pos, item;

   printf("Enter size: ");
   scanf("%d",&n);

   printf("Enter sorted elements:\n");
   for(i=0;i<n;i++)
      scanf("%d",&a[i]);

   printf("Enter element to insert: ");
   scanf("%d",&item);

   for(i=0;i<n;i++)
   {
      if(item<a[i])
      {
         pos=i;
         break;
```

```
        }
    }

    for(i=n;i>pos;i--)
        a[i]=a[i-1];

    a[pos]=item;
    n++;

    printf("Array after insertion:\n");

    for(i=0;i<n;i++)
        printf("%d ",a[i]);

    return 0;
}
```

OUTPUT

Enter size: 5

Enter sorted elements:

10

11

12

14

25

Enter element to insert: 23

Array after insertion:

10 11 12 14 23 25

---

6. Insert an Element in an Unsorted Array

```
#include <stdio.h>

int main()
{
   int a[100], n, i, pos, item;

   printf("Enter size: ");
   scanf("%d",&n);

   printf("Enter elements:\n");
   for(i=0;i<n;i++)
      scanf("%d",&a[i]);

   printf("Enter position: ");
   scanf("%d",&pos);

   printf("Enter element: ");
   scanf("%d",&item);
```

```
    for(i=n;i>=pos;i--)
        a[i]=a[i-1];

    a[pos-1]=item;
    n++;

    printf("Array after insertion:\n");

    for(i=0;i<n;i++)
        printf("%d ",a[i]);

    return 0;
}
```

OUTPUT

Enter size: 4

Enter elements:

1

3

4

45

Enter position: 3

Enter element: 7

Array after insertion:

1 3 7 4 45

7.Matrix Multiplication

```
#include <stdio.h>

int main()
{
   int a[10][10],b[10][10],c[10][10];
   int i,j,k,r1,c1,r2,c2;

   printf("Enter rows and columns of first matrix: ");
   scanf("%d%d",&r1,&c1);

   printf("Enter rows and columns of second matrix: ");
   scanf("%d%d",&r2,&c2);

   if(c1!=r2)
   {
      printf("Matrix multiplication not possible");
      return 0;
   }

   printf("Enter first matrix:\n");
   for(i=0;i<r1;i++)
```

```
    for(j=0;j<c1;j++)
        scanf("%d",&a[i][j]);

printf("Enter second matrix:\n");
for(i=0;i<r2;i++)
    for(j=0;j<c2;j++)
        scanf("%d",&b[i][j]);

for(i=0;i<r1;i++)
{
    for(j=0;j<c2;j++)
    {
        c[i][j]=0;
        for(k=0;k<c1;k++)
            c[i][j]+=a[i][k]*b[k][j];
    }
}

printf("Result Matrix:\n");

for(i=0;i<r1;i++)
{
    for(j=0;j<c2;j++)
        printf("%d ",c[i][j]);
```

```
        printf("\n");
    }


    return 0;
}
```

OUTPUT

Enter rows and columns of first matrix: 2

3

Enter rows and columns of second matrix: 3

2

Enter first matrix:

1

2

3

4

5

6

Enter second matrix:

1

3

4

5

6

7

Result Matrix:

27 34

60 79

8.Matrix Transpose

```
#include <stdio.h>

int main()
{
   int a[10][10], t[10][10];
   int r, c, i, j;

   printf("Enter rows and columns: ");
   scanf("%d%d", &r, &c);

   printf("Enter matrix elements:\n");
   for(i=0;i<r;i++)
      for(j=0;j<c;j++)
         scanf("%d",&a[i][j]);

   for(i=0;i<r;i++)
      for(j=0;j<c;j++)
         t[j][i]=a[i][j];
```

```
    printf("Transpose Matrix:\n");
    for(i=0;i<c;i++)
    {
        for(j=0;j<r;j++)
            printf("%d ",t[i][j]);
        printf("\n");
    }

    return 0;
}
```

OUTPUT

Enter rows and columns: 2

3

Enter matrix elements:

1

13

7

79

7

74

Transpose Matrix:

1 79

13 7

7 74

9.Right Diagonal Sum

```
#include <stdio.h>

int main()
{
   int a[10][10], n, i, j, sum=0;

   printf("Enter order of matrix: ");
   scanf("%d",&n);

   printf("Enter matrix elements:\n");
   for(i=0;i<n;i++)
      for(j=0;j<n;j++)
         scanf("%d",&a[i][j]);

   for(i=0;i<n;i++)
      sum += a[i][n-1-i];

   printf("Right diagonal sum = %d",sum);

   return 0;
```

}

OUTPUT

Enter order of matrix: 2

Enter matrix elements:

1

2

2

2

Right diagonal sum = 4

10.Check Identity Matrix

```
#include <stdio.h>

int main()
{
   int a[10][10], n, i, j, flag=1;

   printf("Enter order of matrix: ");
   scanf("%d",&n);

   printf("Enter matrix elements:\n");
   for(i=0;i<n;i++)
      for(j=0;j<n;j++)
         scanf("%d",&a[i][j]);
```

```
for(i=0;i<n;i++)
{
   for(j=0;j<n;j++)
   {
      if(i==j)
      {
         if(a[i][j]!=1)
            flag=0;
      }
      else
      {
         if(a[i][j]!=0)
            flag=0;
      }
   }
}

if(flag)
   printf("Identity Matrix");
else
   printf("Not an Identity Matrix");

return 0;
```

}

OUTPUT

CASE1:

Enter order of matrix: 3

Enter matrix elements:

1

0

0

0

0

1

0

0

1

Not an Identity Matrix

CASE2:

Enter order of matrix: 2

Enter matrix elements:

1

0

0

1

Identity Matrix

---

11. Count Vowels, Consonants, Digits and White Spaces

```
#include <stdio.h>

int main()
{
   char str[100];
   int i,v=0,c=0,d=0,s=0;

   printf("Enter a string: ");
   fgets(str,sizeof(str),stdin);

   for(i=0;str[i]!='\0';i++)
   {
      if((str[i]>='A'&&str[i]<='Z')||(str[i]>='a'&&str[i]<='z'))
      {
         if(str[i]=='A'||str[i]=='E'||str[i]=='I'||str[i]=='O'||str[i]=='U'||
            str[i]=='a'||str[i]=='e'||str[i]=='i'||str[i]=='o'||str[i]=='u')
```

```
            v++;
        else
            c++;
    }
    else if(str[i]>='0'&&str[i]<='9')
        d++;
    else if(str[i]==' ')
        s++;
  }

  printf("Vowels = %d\n",v);
  printf("Consonants = %d\n",c);
  printf("Digits = %d\n",d);
  printf("White Spaces = %d\n",s);

  return 0;
}
```

OUTPUT

Enter a string: My name is pavi

Vowels = 5

Consonants = 7

Digits = 0

White Spaces = 3

---

12. Frequency of Each Character in a String

```
#include <stdio.h>

int main()
{
   char str[100];
   int i,j,count,len;

   printf("Enter a string: ");
   scanf("%s",str);

   for(len=0;str[len]!='\0';len++);

   for(i=0;i<len;i++)
   {
      count=1;
      if(str[i]!='0')
      {
         for(j=i+1;j<len;j++)
         {
            if(str[i]==str[j])
            {
               count++;
               str[j]='0';
```

```
            }
        }
        printf("%c = %d\n",str[i],count);
    }
  }

  return 0;
}
```

OUTPUT:

Enter a string: pavatharani

p = 1

a = 4

v = 1

t = 1

h = 1

r = 1

n = 1

i = 1

13.Count Words in a String

```
#include <stdio.h>

int main()
```

```
{
   char str[100];
   int i,words=1;

   printf("Enter a sentence: ");
   fgets(str,sizeof(str),stdin);

   for(i=0;str[i]!='\0';i++)
   {
      if(str[i]==' ')
         words++;
   }

   printf("Number of words = %d",words);

   return 0;
}
```

OUTPUT

Enter a sentence: College of enginerring guindy

Number of words = 4

14.Sort String Array

```
#include <stdio.h>
```

```
#include <string.h>

int main()
{
   char str[10][20],temp[20];
   int n,i,j;

   printf("Enter number of strings: ");
   scanf("%d",&n);

   printf("Enter strings:\n");
   for(i=0;i<n;i++)
      scanf("%s",str[i]);

   for(i=0;i<n-1;i++)
   {
      for(j=i+1;j<n;j++)
      {
         if(strcmp(str[i],str[j])>0)
         {
            strcpy(temp,str[i]);
            strcpy(str[i],str[j]);
            strcpy(str[j],temp);
         }
```

```
        }
    }

    printf("Sorted strings:\n");
    for(i=0;i<n;i++)
        printf("%s\n",str[i]);

    return 0;
}
```

OUTPUT

Enter number of strings: 3

Enter strings:

HOW ARE YOU

Sorted strings:

ARE

HOW

YOU

15.Toggle Case of a Sentence

```
#include <stdio.h>

int main()
{
    char str[100];
```

```
    int i;

    printf("Enter a sentence: ");
    fgets(str,sizeof(str),stdin);

    for(i=0;str[i]!='\0';i++)
    {
        if(str[i]>='A'&&str[i]<='Z')
            str[i]=str[i]+32;
        else if(str[i]>='a'&&str[i]<='z')
            str[i]=str[i]-32;
    }

    printf("After Toggle Case:\n%s",str);

    return 0;
}
```

OUTPUT

Enter a sentence: hardwork NEVER fails

After Toggle Case:

HARDWORK never FAILS
